Sufentanil: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Sufentanil: Patient drug information” and ”Sufentanil: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates   August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term

ALERT: US Boxed Warning

Accidental exposure and Risk Evaluation and Mitigation Strategy (REMS) program (sublingual tablet):

Accidental exposure to or ingestion of sufentanil, especially in children, can result in respiratory depression and death. Because of the potential for life-threatening respiratory depression due to accidental exposure, sufentanil is only available through a restricted program called the Dsuvia REMS Program. Sufentanil must only be dispensed to patients in a certified medically supervised health care setting. Discontinue use of sufentanil prior to discharge or transfer from the certified medically supervised health care setting.

Life-threatening respiratory depression:

Serious, life-threatening, or fatal respiratory depression may occur with use of sufentanil, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of sufentanil are essential.

Addiction, abuse, and misuse:

Because the use of sufentanil exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing sufentanil, and reassess all patients regularly for the development of these behaviors and conditions.

Cytochrome P450 3A4 interaction:

The concomitant use of sufentanil with all cytochrome P450 3A4 inhibitors may result in an increase in sufentanil plasma concentrations, which could increase or prolong adverse drug reactions and may cause potentially fatal respiratory depression. In addition, discontinuation of a concomitantly used cytochrome P450 3A4 inducer may result in an increase in sufentanil plasma concentration. Monitor patients receiving sufentanil and any CYP3A4 inhibitor or inducer.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other central nervous system (CNS) depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of sufentanil and benzodiazepines or other CNS depressants for use in patients for whom alternative treatment options are inadequate.

Brand Names: US

- Dsuvia

Pharmacologic Category

- Analgesic, Opioid;
- Anilidopiperidine Opioid;
- General Anesthetic

Dosing: Adult

Dosage guidance:

Dosing: Dosing should be individualized based on patient-specific factors (eg, comorbidities, severity of pain, degree of opioid experience/tolerance) and titrated to patient-specific treatment goals (eg, improvement in function and quality of life, decrease in pain using a validated pain rating scale). Use the lowest effective dose for the shortest period of time.

Clinical considerations: Opioids may be part of a comprehensive, multimodal, patient-specific treatment plan for managing moderate to severe pain. Maximize nonopioid analgesia (when appropriate) prior to initiation of opioid analgesia (Ref).

Acute pain management

Acute pain management: Sublingual tablet: Initial: Sublingual: 30 mcg; may repeat as needed with a minimum of 1 hour between doses; maximum dose: 360 mcg/day (12 tablets); do not use for >72 hours.

Note: Only administer in a certified medically supervised health care setting by a health care provider; discontinue treatment prior to discharge from supervised setting.

Analgesia, surgical

Analgesia, surgical:

Surgical analgesia (as a component of balanced anesthesia) (surgery expected to last: 1 to 2 hours): IV: Total dose: 1 to 2 mcg/kg with N_2O/O_2; ≥75% of total dose may be administered by slow injection or infusion prior to intubation (titrate to individual response).

Maintenance:

Incremental dosing: According to the manufacturer, 10 to 25 mcg may be administered as needed when movement and/or changes in vital signs indicate surgical stress or lightening of analgesia. May also administer doses in the range of 5 to 20 mcg as needed (Ref) **or** 0.1 to 0.25 mcg/**kg** as needed (Ref). Total dose should not exceed 1 mcg/kg/hour of expected surgical time.

Continuous infusion: May also be administered as a continuous infusion with the infusion rate based on the induction dose used. Maximum infusion rate according to the manufacturer: 1 mcg/kg/hour. May also administer doses in the range of 0.3 to 0.9 mcg/kg/hour (Ref) **or** 0.5 to 1.5 mcg/kg/hour (Ref).

Surgical analgesia (as a component of balanced anesthesia) (surgery expected to last 2 to 8 hours): Total dose: 2 to 8 mcg/kg with N_2O/O_2; ≤75% of total dose may be administered by slow injection or infusion prior to intubation (titrate to individual response).

Maintenance:

Incremental dosing: According to the manufacturer, 10 to 50 mcg may be administered as needed when movement and/or changes in vital signs indicate surgical stress or lightening of analgesia. Total dose should not exceed 1 mcg/kg/hour of expected surgical time.

Continuous infusion: May also be administered as a continuous infusion with the infusion rate based on the induction dose used. Maximum infusion rate according to the manufacturer: 1 mcg/kg/hour. May also administer doses in the range of 0.3 to 0.9 mcg/kg/hour (Ref) **or** 0.5 to 1.5 mcg/kg/hour (Ref).

Anesthesia, surgical

Anesthesia, surgical: Total dose: 8 to 30 mcg/kg as a slow injection, infusion, or injection followed by infusion; titrate to individual patient response. **Note:** In patients administered high doses of sufentanil, qualified personnel and adequate facilities are necessary to manage the potential for postoperative respiratory depression.

Maintenance:

Incremental dosing: 0.5 to 10 mcg/kg as needed in anticipation of surgical stress

Continuous infusion: Base infusion rate on the induction dose so that the total dose for the procedure does not exceed 30 mcg/kg

Analgesia for labor and delivery

Analgesia for labor and delivery: Epidural: 10 to 15 mcg with bupivacaine 0.125% with/without epinephrine. Dose can be repeated twice (for a total of 3 doses) at not less than 1-hour intervals until delivery.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion in the urine (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution and reduce dose as needed; titrate slowly and closely monitor for signs of CNS and respiratory depression.

Dosing: Obesity: Adult

IV: In adult obese patients (eg, >20% above ideal body weight), use lean body weight to determine dosage.

Dosing: Older Adult

Refer to adult dosing. Use with caution and titrate slowly; monitor closely for signs of CNS and respiratory depression.

Dosing: Pediatric

(For additional information see "Sufentanil: Pediatric drug information")

Note: Doses should be titrated to appropriate effects; wide range of doses, dependent upon patient age (particularly young infants), desired degree of analgesia or anesthesia; use lean body weight to dose patients who are >20% above ideal body weight. Should only be administered under the supervision of a qualified physician experienced in the use of anesthetics.

Anesthesia, cardiac surgery

Anesthesia, cardiac surgery: Infants, Children, and Adolescents: IV: Initial: 5 to 25 mcg/**kg**; repeat maintenance doses: 1 to 5 mcg/**kg**/doses up to 25 to 50 mcg/dose as needed based on response to initial dose and as determined by changes in vital signs indicating surgical stress or lightening of anesthesia; data based on experience in cardiac surgery, which requires much higher induction and maintenance doses; lower dosing may be sufficient for other procedures (Ref).

Epidural

Epidural: Limited data available: Infants ≥3 months and Children: Epidural (preservative free) injection: Initial bolus: 0.2 mcg/kg followed by continuous infusion at 0.1 mcg/kg/hour in combination with ropivacaine (Ref).

Sedation, preoperative or procedural

Sedation, preoperative or procedural: Limited data available: Children ≥3 years: Intranasal (using parenteral preparation): 1 to 3 mcg/kg in combination with other agents (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution and reduce dose as needed; titrate slowly and closely monitor for signs of CNS and respiratory depression. A case-series evaluation of six pediatric patients (age range: 10 to 15 years) with chronic renal failure undergoing transplantation did not show overall differences in pharmacokinetic parameters compared to normal renal function; however, amongst the pediatric patients, a high degree of variability was observed; dosing should be individualized (Ref)

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution and reduce dose as needed; titrate slowly and closely monitor for signs of CNS and respiratory depression.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Dermatologic: Pruritus (epidural administration with bupivacaine: 25%)

Gastrointestinal: Nausea (29%)

Nervous system: Headache (12%)

1% to 10%:

Cardiovascular: Hypotension (5%)

Gastrointestinal: Vomiting (6%)

Nervous system: Dizziness (6%)

Frequency not defined:

Cardiovascular: Bradycardia, ECG abnormality, flushing, hypertension, orthostatic hypotension, peripheral vasodilation, presyncope, sinus tachycardia, syncope

Dermatologic: Hyperhidrosis, skin rash

Gastrointestinal: Abdominal distention, abdominal distress, constipation, decreased gastrointestinal motility, diarrhea, dyspepsia, eructation, flatulence, gastritis, hiccups, intestinal obstruction (postoperative), oral hypoesthesia, retching, upper abdominal pain, xerostomia

Genitourinary: Decreased urine output, oliguria, urinary hesitancy, urinary retention

Hepatic: Increased liver enzymes, increased serum aspartate aminotransferase

Nervous system: Agitation, anxiety, confusion, disorientation, drowsiness, drug abuse, euphoria, hallucination, hemiparesis, insomnia, lethargy, memory impairment, mental status changes, opioid dependence, sedated state

Neuromuscular & skeletal: Muscle rigidity, muscle spasm

Ophthalmic: Miosis

Renal: Kidney failure

Respiratory: Apnea, atelectasis, bradypnea, hypoventilation, hypoxia, oxygen saturation decreased, respiratory depression, respiratory distress, respiratory failure

Postmarketing:

Hypersensitivity: Anaphylaxis

Nervous system: Allodynia (opioid-induced hyperalgesia) (Ref)

Contraindications

Hypersensitivity (eg, anaphylaxis) to sufentanil or any component of the formulation

Additional contraindications for sublingual tablets: Significant respiratory depression; acute or severe bronchial asthma in an unmonitored setting or in the absence of resuscitative equipment; known or suspected gastrointestinal obstruction, including paralytic ileus

Canadian labeling: Additional contraindications (not in US labeling): Hypersensitivity to fentanyl or other morphinomimetics; IV use during labor or before clamping cord during cesarean section; epidural administration in patients with severe hemorrhage or shock, septicemia, or infection at proposed puncture site

Warnings/Precautions

Concerns related to adverse effects:

• Adrenocortical insufficiency: Use with caution in patients with adrenal insufficiency, including Addison disease. Long-term opioid use may cause secondary hypogonadism, which may lead to sexual dysfunction, infertility, mood disorders, and osteoporosis (Brennan 2013).

• Bradyarrhythmias: Severe bradycardia may occur; use with caution in patients with bradyarrhythmias.

• CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery, driving).

• Esophageal dysfunction: Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.

• Hyperalgesia: Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.

• Hypotension: May cause severe hypotension (including orthostatic hypotension and syncope); use with caution in patients with hypovolemia, cardiovascular disease (including acute myocardial infarction), or drugs that may exaggerate hypotensive effects (including phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration. Use injection with caution in patients with circulatory shock. Avoid use of sublingual tablets in patients with circulatory shock.

• Respiratory depression: Fatal respiratory depression may occur. Carbon dioxide retention from opioid-induced respiratory depression can exacerbate the sedating effects of opioids. Patients and caregivers should be educated on how to recognize respiratory depression and the importance of getting emergency assistance immediately (eg, calling 911) in the event of known or suspected overdose.

• Serotonin syndrome: Potentially life-threatening serotonin syndrome (SS) has occurred with concomitant use of sufentanil and serotonergic agents (eg, selective serotonin reuptake inhibitors, serotonin–norepinephrine reuptake inhibitors, triptans, tricyclic antidepressants, 5-HT receptor antagonists, mirtazapine, trazodone, tramadol) and agents that impair metabolism of serotonin (eg, monoamine oxidase inhibitors). Monitor patients closely for signs of SS such as mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile BP, diaphoresis); neuromuscular changes (eg, tremor, rigidity, myoclonus); GI symptoms (eg, nausea, vomiting, diarrhea); and/or seizures. Discontinue sufentanil if serotonin syndrome is suspected.

Disease-related concerns:

• Abdominal conditions: May obscure diagnosis or clinical course of patients with acute abdominal conditions. Use of sublingual tablets is contraindicated with known or suspected GI obstruction, including paralytic ileus.

• Biliary tract impairment: Use with caution in patients with biliary tract dysfunction, including acute pancreatitis; opioids may cause constriction of sphincter of Oddi.

• CNS depression/coma: Use with caution in patients with impaired consciousness or coma as these patients are susceptible to intracranial effects of CO_2 retention.

• Delirium tremens: Use opioids with caution in patients with delirium tremens.

• Head trauma: Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur.

• Hepatic impairment: Use with caution and reduce dose as needed in patients with hepatic impairment.

• Obesity: Use with caution in patients who are morbidly obese. Consider use of lean body weight for dosing in patients >20% over ideal body weight.

• Prostatic hyperplasia/urinary stricture: Use opioids with caution in patients with prostatic hyperplasia and/or urinary stricture.

• Psychosis: Use opioids with caution in patients with toxic psychosis.

• Renal impairment: Use with caution and reduce dose as needed in patients with renal impairment.

• Respiratory disease: Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale, and those with a substantially decreased respiratory reserve, hypoxia, hypercapnia, or preexisting respiratory depression, particularly when initiating and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages.

• Seizures: Use with caution in patients with a history of seizure disorders; may increase risk or exacerbate preexisting seizure disorders.

• Sleep-related disorders: Use with caution in patients with sleep-related disorders, including sleep apnea, due to increased risk for respiratory and CNS depression. Monitor carefully and titrate dosage cautiously in patients with mild sleep-disordered breathing. Avoid opioids in patients with moderate to severe sleep-disordered breathing (CDC [Dowell 2022]).

• Thyroid dysfunction: Use opioids with caution in patients with thyroid dysfunction.

Concurrent drug therapy issues:

• Benzodiazepines or other CNS depressants: Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in hypotension, profound sedation, respiratory depression, coma, and death. Following the administration of sufentanil, the dose of other CNS depressant drugs should be reduced. Consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmefene) for emergency treatment of opioid overdose in patients taking benzodiazepines or other CNS depressants concomitantly with opioids.

• CYP3A4 interactions: Use with all CYP3A4 inhibitors may result in an increase in sufentanil plasma concentrations, which could increase or prolong adverse drug effects and may cause potentially fatal respiratory depression. In addition, discontinuation of a concomitant CYP3A4 inducer may result in increased sufentanil concentrations. Monitor patients receiving sufentanil and any CYP3A4 inhibitor or inducer.

Special populations:

• Cachectic or debilitated patients: Use with caution in cachectic or debilitated patients; there is a greater potential for critical respiratory depression, even at therapeutic dosages.

• Older adult: Use opioids with caution in older adults; may be more sensitive to adverse effects. Clearance may also be reduced in older adults (with or without renal impairment) resulting in a narrow therapeutic window and increased adverse effects. Monitor closely for adverse effects associated with opioid therapy (eg, respiratory and CNS depression, falls, cognitive impairment, constipation) (CDC [Dowell 2022]). Consider the use of alternative nonopioid analgesics in these patients when possible.

• Neonates: Neonatal opioid withdrawal syndrome: Signs and symptoms include irritability, hyperactivity, and abnormal sleep pattern, high pitched cry, tremor, vomiting, diarrhea, and failure to gain weight. Onset, duration, and severity depend on the drug used, duration of use, maternal dose, and rate of drug elimination by the newborn. If opioid use is required for a prolonged period in a pregnant woman, advise the patient of the risk of neonatal opioid withdrawal syndrome and ensure that appropriate treatment will be available.

• Pediatric: Use caution in neonates as clearance of sufentanil is much slower than adults; neonates with cardiovascular disease have an even slower clearance. The clearance of sufentanil in children 2 to 8 years of age was shown to be twice as rapid as that seen in adults and adolescents (Lundeberg 2011).

Other warnings/precautions:

• Abuse/misuse/diversion: Use with caution in patients with a history of substance abuse disorder; potential for drug dependency exists. Other factors associated with increased risk for misuse include concomitant depression or other mental health conditions, higher opioid dosages, long-term use, or taking other CNS depressants. Consider offering an opioid overdose reversal agent (eg, naloxone, nalmefene) prescription in patients with an increased risk for overdose, such as history of overdose or substance use disorder, higher opioid dosages (≥50 morphine milligram equivalents/day orally), concomitant benzodiazepine use, and patients at risk for returning to a high dose after losing tolerance (CDC [Dowell 2022]).

• Accidental exposure (sublingual tablet): Accidental exposure to or ingestion of sufentanil, especially in children, can result in respiratory depression and death.

• Appropriate use: Epidural: Proper placement of the needle or catheter in the epidural space should be verified before injection to assure that unintentional intravascular or intrathecal administration does not occur. Unintentional intravascular injection may result in a potentially serious overdose, including acute truncal muscular rigidity and apnea. Unintentional intrathecal injection of the full sufentanil/bupivacaine epidural doses and volume may produce effects of high spinal anesthesia, including prolonged paralysis and delayed recovery.

• Opioid overdose reversal agent: Discuss the availability of opioid overdose reversal agents (eg, naloxone, nalmefene) with all patients who are prescribed opioid analgesics, as well as their caregivers, and consider prescribing it to patients who are at increased risk of opioid overdose. These include patients who are also taking benzodiazepines or other CNS depressants, have an opioid use disorder (OUD) (current or history of), or have experienced opioid-induced respiratory depression/opioid overdose. Additionally, health care providers should consider prescribing an opioid overdose reversal agent to patients prescribed medications to treat OUD; patients at risk of opioid overdose even if they are not taking an opioid analgesic or medication to treat OUD; and patients taking opioids, including methadone or buprenorphine for OUD, if they have household members, including children, or other close contacts at risk for accidental ingestion or opioid overdose. Inform patients and caregivers on options for obtaining an opioid overdose reversal agent (eg, by prescription, OTC, directly from a pharmacist, a community-based program) as permitted by state dispensing and prescribing guidelines. Educate patients and caregivers on how to recognize respiratory depression, proper administration and important differences of opioid overdose reversal agents, and getting emergency help even if an opioid overdose reversal agent is administered.

• Rapid infusion: Injection: Inject slowly over at least 2 minutes (Sebel 1982); rapid IV infusion may result in skeletal muscle and chest wall rigidity, impaired ventilation, or respiratory distress/arrest; nondepolarizing skeletal muscle relaxant may be required.

• Trained individuals: Injection: Sufentanil should be administered by health care providers specifically trained in the use of anesthetic agents and should not be used in diagnostic or therapeutic procedures outside the monitored anesthesia setting; resuscitative and intubation equipment should be readily available.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous [preservative free]:

Generic: 50 mcg/mL (1 mL); 100 mcg/2 mL (2 mL [DSC]); 250 mcg/5 mL (5 mL [DSC])

Tablet Sublingual, Sublingual, as citrate:

Dsuvia: 30 mcg [contains fd&c blue #2 (indigotine,indigo carmine)]

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (SUFentanil Citrate Intravenous)

50 mcg/mL (per mL): $23.45

Sublingual (Dsuvia Sublingual)

30 mcg (per each): $96.00

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 50 mcg/mL (1 mL, 5 mL)

Controlled Substance

C-II

Administration: Adult

IV: Intermittent doses may be administered IV as either a slow injection (eg, over at least 2 minutes (Ref) or a range of 2 to 10 minutes (Ref)) or as an infusion. May also be administered as a continuous infusion.

Oral: Sublingual tablet: Administer by a health care provider in a certified medically supervised health care setting only. Wear gloves when administering. Tablet is packaged in a single-dose applicator; do not open until ready to administer. Patient should open their mouth and touch their tongue to roof of mouth. Using the single-dose applicator, administer into sublingual space. Patient should not chew or swallow tablet; do not eat or drink and minimize talking for 10 minutes after administration. Refer to manufacturer's labeling for additional information.

Administration: Pediatric

Parenteral: IV: Administer at a concentration ≤ 50 mcg/mL (undiluted)

Intermittent IV: May be administered by either slow injection or as an infusion.

Slow IV injection: Administer over at least 2 minutes (Ref); in adults, a range of 2 to 10 minutes has been used (Ref).

Continuous IV infusion: Administer as a continuous infusion via an infusion pump

Intranasal: Administer using a needleless syringe over 15 to 20 seconds (Ref) or by a nasal actuating device (eg, GoMedical Nasal Device or MAD nasal drug delivery device) (Ref)

Preparation for Administration: Pediatric

IV: May further dilute in an appropriate fluid to a concentration <50 mcg/mL

Storage/Stability

Injection: Store at 20°C to 25°C (68°F to 77°F). Protect from light.

Sublingual tablet: Store at 20°C to 25°C (68°F to 77°F); excursions permitted at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Acute pain management (sublingual tablet): Management of acute pain severe enough to require an opioid analgesic and for which alternative treatments are inadequate.

Limitations of use: Because of the risks of substance use disorder, abuse, misuse, overdose, and death, which can occur at any dosage or duration and persist over the course of therapy, reserve opioid analgesics, including sufentanil, for use in patients for whom alternative treatment options are ineffective, not tolerated, or would be otherwise inadequate to provide sufficient management of pain. Administer by health care provider in a health care setting (eg, hospital, surgical center, emergency department) only; not for home use. Do not administer >72 hours (has not been studied).

Epidural analgesia (injection): For epidural administration as an analgesic combined with low-dose bupivacaine during labor and vaginal delivery.

Surgical analgesia (injection): Analgesic adjunct for the maintenance of balanced general anesthesia in patients who are intubated and ventilated.

Surgical anesthesia (injection): As a primary anesthetic agent for the induction and maintenance of anesthesia with 100% oxygen in patients undergoing major surgical procedures; in patients who are intubated and ventilated, such as cardiovascular surgery or neurosurgical procedures in the sitting position; to provide favorable myocardial and cerebral oxygen balance or when extended postoperative ventilation is anticipated.

Medication Safety Issues

Sound-alike/look-alike issues:

SUFentanil may be confused with ALfentanil, fentaNYL

Sufenta may be confused with Alfenta, Sudafed, Survanta

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV; epidural and intrathecal medications; opioids, all formulations and routes of administrations) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Metabolism/Transport Effects

Substrate of CYP3A4 (Major with inhibitors), CYP3A4 (Minor with inducers); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ceritinib: SUFentanil may increase bradycardic effects of Ceritinib. Ceritinib may increase serum concentrations of SUFentanil. Management: If ceritinib is initiated in a patient on sufentanil, consider a sufentanil dose reduction and monitor for increased sufentanil effects and toxicities (eg, respiratory depression). Also monitor for symptomatic bradycardia if these agents are combined. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase analgesic effects of Opioid Agonists. Cimetidine may increase respiratory depressant effects of Opioid Agonists. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible.

These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of SUFentanil. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of SUFentanil. Management: If a strong CYP3A4 inducer is initiated in a patient on sufentanil, consider a sufentanil dose increase and monitor for decreased sufentanil effects and opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of SUFentanil. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of SUFentanil. Management: If a strong CYP3A4 inhibitor is initiated in a patient on sufentanil, consider a sufentanil dose reduction and monitor for increased sufentanil effects and toxicities (eg, respiratory depression). *Risk D: Consider Therapy Modification*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Eluxadoline: Opioid Agonists may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: SUFentanil may increase adverse/toxic effects of Monoamine Oxidase Inhibitors. Specifically, the risk for serotonin syndrome or opioid toxicities (eg, respiratory depression, coma) may be increased. Management: Sufentanil should not be used with monoamine oxidase (MAO) inhibitors (or within 14 days of stopping an MAO inhibitor) due to the potential for serotonin syndrome and/or excessive CNS depression. *Risk X: Avoid*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nefazodone: Opioid Agonists (metabolized by CYP3A4) may increase serotonergic effects of Nefazodone. This could result in serotonin syndrome. Nefazodone may increase serum concentrations of Opioid Agonists (metabolized by CYP3A4). Management: If concomitant use of opioid agonists that are metabolized by CYP3A4 and nefazodone is necessary, consider dose reduction of the opioid until stable drug effects are achieved. Monitor for increased opioid effects and serotonin syndrome/serotonin toxicity. *Risk D: Consider Therapy Modification*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid agonists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

PHENobarbital: May increase CNS depressant effects of SUFentanil. PHENobarbital may decrease serum concentrations of SUFentanil. Management: Avoid use of sufentanil and phenobarbital when possible. Monitor for respiratory depression/sedation. Because phenobarbital is also a moderate CYP3A4 inducer, monitor for decreased sufentanil efficacy and withdrawal if combined. *Risk D: Consider Therapy Modification*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Primidone: May increase CNS depressant effects of SUFentanil. Primidone may decrease serum concentrations of SUFentanil. Management: Avoid use of sufentanil and primidone when possible. Monitor for respiratory depression/sedation. Because primidone is also a moderate CYP3A4 inducer, monitor for decreased sufentanil efficacy and withdrawal if combined. *Risk D: Consider Therapy Modification*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Serotonergic Agents (High Risk): Opioid Agonists (metabolized by CYP3A4) may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Chronic opioid use may cause hypogonadism and hyperprolactinemia, which may decrease fertility in patients of reproductive potential. Menstrual cycle disorders (including amenorrhea), erectile dysfunction, and impotence have been reported. The incidence of hypogonadism may be increased with the use of opioids in high doses or long-acting opioid formulations. It is not known if the effects on fertility are reversible. Monitor patients on long-term therapy (de Vries 2020; Gadelha 2022). Sufentanil is not indicated for chronic use.

Pregnancy Considerations

Sufentanil crosses the placenta (Cuypers 1995; Loftus 1995).

Maternal use of opioids may be associated with poor fetal growth, stillbirth, and preterm delivery (CDC [Dowell 2022]). Opioids used as part of obstetric analgesia/anesthesia during labor and delivery may temporarily affect the fetal heart rate (ACOG 2019b).

Neonatal abstinence syndrome (NAS)/neonatal opioid withdrawal syndrome (NOWS) may occur following prolonged in utero exposure to opioids (CDC [Dowell 2022]). NAS/NOWS may be life-threatening if not recognized and treated and requires management according to protocols developed by neonatology experts. Presentation of symptoms varies by opioid characteristics (eg, immediate release, sustained release), time of last dose prior to delivery, drug metabolism (maternal, placental, and infant), net placental transfer, as well as other factors (AAP [Hudak 2012]; AAP [Patrick 2020]). Clinical signs characteristic of withdrawal following in utero opioid exposure include excessive crying or easily irritable, fragmented sleep (<2 to 3 hours after feeding), tremors, increased muscle tone, or GI dysfunction (hyperphagia, poor feeding, feeding intolerance, watery or loose stools) (Jilani 2022). NAS/NOWS occurs following chronic opioid exposure and would not be expected following the use of opioids at delivery (AAP [Patrick 2020]).

Monitor infants of mothers on long-term/chronic opioid therapy for symptoms of withdrawal. Symptom onset reflects the half-life of the opioid used. Monitor infants for at least 3 days following exposure to immediate-release opioids; monitor for at least 4 to 7 days following exposure to sustained-release opioids (AAP [Patrick 2020]; CDC [Dowell 2022]). Monitor newborns for excess sedation and respiratory depression when opioids are used during labor.

When opioids are needed to treat acute pain in pregnant patients, the lowest effective dose for only the expected duration of pain should be prescribed (CDC [Dowell 2022]).

Sufentanil injection is commonly used to treat maternal pain during labor and immediately postpartum (ACOG 2019b). Administration of epidural sufentanil with bupivacaine with or without epinephrine is indicated for use in labor and delivery. Sufentanil tablets are not recommended for use during or immediately before labor. Opioid use for pain following vaginal or cesarean delivery should be made as part of a shared decision-making process. A stepwise multimodal approach to managing postpartum pain is recommended. A low-dose, low-potency, short-acting opioid can be used to treat acute pain associated with delivery when needed (ACOG 2021).

Opioids are not preferred for the treatment of chronic noncancer pain during pregnancy; consider strategies to minimize or avoid opioid use (ACOG 2017; CDC [Dowell 2022]). Sufentanil is not indicated for chronic use.

Sufentanil injection is approved for use in surgical procedures. Pregnant patients should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 2019a).

Breastfeeding Considerations

Sufentanil is present in breast milk.

Sufentanil concentrations in breast milk were evaluated in 29 women undergoing elective cesarean delivery. All women received epidural sufentanil 20 mcg with bupivacaine; 11 women also received additional sufentanil via patient controlled epidural analgesia (PCEA). Breast milk was sampled on days 1, 2, and 3 following surgery. Sufentanil was detected in breast milk obtained on all 3 days; however, not all women provided samples and sufentanil was not present in every sample obtained. Sufentanil breast milk concentrations were greater than those in the maternal plasma in all samples. The highest concentrations were on day 3 in women using PCEA. Adverse events were not observed in the newborns (Cuypers 1995). Sufentanil was not detected in the colostrum of nine women who received a single epidural dose of 50 mcg prior to elective cesarean delivery; sampling occurred ~1 hour after sufentanil administration (Madej 1987).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother. Nonopioid analgesics are preferred for lactating patients who require pain control peripartum or for surgery outside of the postpartum period. When opioids are needed for lactating patients, use the lowest effective dose for the shortest duration of time to limit adverse events in the mother and breastfeeding infant. Analgesics administered by the epidural route help limit infant exposure (AAP [Sachs 2013]; ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; WHO 2002). Sufentanil is not indicated for chronic use.

Monitor infants exposed to opioids via breast milk for drowsiness, sedation, feeding difficulties, or limpness (ACOG 2019b). Withdrawal symptoms may occur when maternal use is discontinued, or breastfeeding is stopped.

The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Pain relief; respiratory and cardiovascular status; BP and heart rate; signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [noncardiac], dysphagia, regurgitation).

Mechanism of Action

Binds to opioid receptors throughout the CNS. Once receptor binding occurs, effects are exerted by opening K+ channels and inhibiting Ca++ channels. These mechanisms increase pain threshold, alter pain perception, inhibit ascending pain pathways; short-acting opioid; dose-related inhibition of catecholamine release (up to 30 mcg/kg) controls sympathetic response to surgical stress.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Analgesia: IV: 1 to 3 minutes; Epidural: 10 minutes; Sublingual tablets: ~30 minutes (Fisher 2018).

Duration: Dose dependent: Anesthesia adjunct doses:

IV: Context-sensitive half-time and recovery varies depending on dose, duration of administration, concomitant anesthesia, and other clinical factors (Miller 2015).

Epidural: 10 to 15 mcg with bupivacaine 0.125%: 1.7 hours.

Sublingual tablets: ~3 hours (Fisher 2018).

Distribution: V_{dss}: Children 2 to 8 years of age: 2.9 ± 0.6 L/kg; Adults: 1.7 ± 0.2 L/kg.

Protein binding: Neonates: 79%; Adults: 91% to 93%; primarily to alpha 1-acid glycoprotein.

Metabolism: Primarily hepatic and small intestine via demethylation and dealkylation.

Bioavailability: Sublingual tablet: ~53%.

Half-life elimination:

IV: Neonates: 7.2 ± 2.7 hours; Infants and Children (2 to 8 years of age): 97 ± 42 minutes; Adolescents 10 to 15 years of age: 76 ± 33 minutes; Adults: 164 minutes.

Sublingual tablet: 2.5 ± 0.85 hours (Fisher 2018).

Time to peak: Sublingual tablet: 1 hour.

Excretion: Urine (2% excreted as unchanged drug; 80% metabolites) within 24 hours.

Clearance:

Children 2 to 8 years: 30.5 ± 8.8 mL/minute/kg.

Adolescents: 12.8 ± 12 mL/minute/kg.

Adults: 12.7 ± 0.8 mL/minute/kg.

Patient Counseling Points

(For additional information see ”Sufentanil: Patient drug information”)

What is this drug used for?

• It is used to manage pain when non-opioid pain drugs do not treat your pain well enough or you cannot take them.

• It is used to put you to sleep for surgery.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Upset stomach or throwing up

• Feeling dizzy or sleepy

• Headache

• Itching

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating

• Severe dizziness or passing out

• Seizures

• Feeling confused

• Trouble breathing, slow breathing, or shallow breathing

• Noisy breathing

• Breathing problems during sleep (sleep apnea)

• Chest pain or pressure

• Fast or slow heartbeat

• Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache

• Adrenal gland problem like feeling very tired or weak, you pass out, you have severe dizziness, very upset stomach, throwing up, or decreased appetite

• Lowered interest in sex, fertility problems, no menstrual period, or ejaculation problems

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AT) Austria: Sufenta | Sufentanil Actavis | Sufentanil hameln | Sufentanil nycomed | Sufentanil torrex pharma | Zalviso;
- (BE) Belgium: Dzuveo | Sufentanil | Zalviso;
- (BG) Bulgaria: Sufentanil torrex;
- (CZ) Czech Republic: Sufenta;
- (DE) Germany: Dzuveo | Sufentanil Actavis | Sufentanil hameln | Sufentanil Hikma | Zalviso;
- (DK) Denmark: Sufentanil hameln;
- (ES) Spain: Zalviso;
- (FI) Finland: Dzuveo | Sufentanil hameln | Sufentanil Narco-med;
- (FR) France: Dzuveo | Zalviso;
- (GB) United Kingdom: Dzuveo | Zalviso;
- (IE) Ireland: Dzuveo | Zalviso;
- (IT) Italy: Dzuveo | Fentatienil | Zalviso;
- (KR) Korea, Republic of: Hana sufentanil | Hanlim Sufentanil | Sufental;
- (LT) Lithuania: Sufenta;
- (LU) Luxembourg: Sufenta;
- (LV) Latvia: Sufenta;
- (MX) Mexico: Zuftil;
- (NL) Netherlands: Dzuveo | Zalviso;
- (NO) Norway: Dzuveo | Sufentanil hameln;
- (PL) Poland: Sufenta;
- (PR) Puerto Rico: Dsuvia | Sufenta;
- (PT) Portugal: Sufentanilo | Zalviso;
- (SE) Sweden: Dzuveo;
- (SI) Slovenia: Sufenta;
- (SK) Slovakia: Sufenta;
- (TW) Taiwan: Sufenta;
- (ZA) South Africa: Micro sufentanil | Pharma q sufentanil | Sufenta | Sufentanil

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